Statistical Analysis Report: MANOVA of Training Methods and Learning Outcomes

Evaluation of Live vs. Recorded Training Delivery on Design Modelling and Technical Expertise

Methodology: One-Way MANOVA (IBM SPSS Statistics)	**Dataset:** manova (1).sav (N = 330)
Status: Completed – Portfolio Documentation	**Design:** Balanced (110 per condition)

1. Executive Summary

This report presents a statistical analysis examining whether different training delivery methods are associated with differences in two learning outcomes: **Design Modelling** and **Technical Expertise**.

The analysis was conducted using IBM SPSS Statistics and a dataset containing **330 observations** distributed equally across three training conditions:

- **Trainer 1:** Live training delivered by Trainer 1
- **Trainer 2:** Live training delivered by Trainer 2
- **Recorded Sessions by Trainer 1:** Recorded training sessions delivered by Trainer 1

Each training group contained 110 observations, resulting in a balanced research design.

A **one-way Multivariate Analysis of Variance (MANOVA)** was conducted to evaluate whether the training condition had a statistically significant effect on the combined outcomes of Design Modelling and Technical Expertise. The overall MANOVA was statistically significant, indicating that the combined performance outcomes differed across the three training conditions.

Follow-up univariate ANOVAs showed statistically significant differences between training groups for both dependent variables. However, the effect was considerably stronger for **Technical Expertise** than for **Design Modelling**.

Tukey HSD post-hoc comparisons showed that participants receiving live instruction from **Trainer 1** achieved significantly higher Technical Expertise scores than both Trainer 2 and the Recorded Sessions condition. Trainer 1 also produced significantly higher Design Modelling scores than the Recorded Sessions condition.

> **Core Finding**
> Overall, the results indicate that the **Trainer 1 live training condition produced the strongest observed performance outcomes**, particularly in Technical Expertise.

2. Research Context

The purpose of this analysis was to examine whether the method used to deliver training was associated with differences in participant performance.

The study compares three training conditions:

1. **Trainer 1** – Live training delivered by Trainer 1
2. **Trainer 2** – Live training delivered by Trainer 2
3. **Recorded Sessions by Trainer 1** – Recorded training sessions delivered by Trainer 1

The analysis evaluates performance across two outcome measures: **Design Modelling** and **Technical Expertise**. Because the study evaluates multiple dependent variables simultaneously, MANOVA was selected as the primary inferential statistical method.

The analysis was conducted as part of a hands-on learning and applied research project using IBM SPSS Statistics.

3. Research Objectives

The analysis was designed to address the following objectives:

- To examine the descriptive performance characteristics of participants across the three training conditions.
- To determine whether training condition has a statistically significant multivariate effect on the combined outcomes of Design Modelling and Technical Expertise.
- To determine whether training condition has statistically significant individual effects on Design Modelling and Technical Expertise.
- To identify which specific training groups differ from one another using Tukey HSD post-hoc comparisons.
- To compare the relative magnitude of the training effect across the two learning outcomes.
- To interpret the practical implications of the statistical findings for training and development.

4. Data and Research Design

The original dataset used for the analysis was the SPSS data file: `manova (1).sav`. The dataset contains **330 observations**.

The independent variable was **Training_Group**. This categorical factor contained three groups:

Training Group	Description	Sample Size
Trainer 1	Live training by Trainer 1	110
Trainer 2	Live training by Trainer 2	110
Recorded Sessions by Trainer 1	Recorded training sessions	110
Total	**Balanced Design**	**330**

The research design is balanced because each training group contains the same number of observations.

5. Variables

5.1 Independent Variable

Training_Group: Represents the training delivery condition. It consists of three categories: Trainer 1, Trainer 2, and Recorded Sessions by Trainer 1.

5.2 Dependent Variables

Design_Modelling: Represents one of the measured learning outcomes related to design and modelling performance.

Technical_Expertise: Represents the second measured learning outcome related to technical performance or domain expertise.

Both variables were treated as continuous dependent variables in the MANOVA.

6. Descriptive Statistics

Descriptive statistics were examined for each training group before evaluating the inferential statistical results.

6.1 Design Modelling

Training Group	N	Mean	Standard Deviation
Trainer 1	110	6.1909	1.8196
Trainer 2	110	5.5818	2.3316
Recorded Sessions by Trainer 1	110	5.3727	1.6848
Overall	**330**	**5.7152**	**1.9898**

Trainer 1 recorded the highest mean Design Modelling score at **6.1909**. Trainer 2 recorded a mean of **5.5818**, while the Recorded Sessions group had the lowest mean of **5.3727**.

The descriptive results therefore suggest a performance advantage for Trainer 1, although the differences between the groups are relatively modest compared with those observed for Technical Expertise.

6.2 Technical Expertise

Training Group	N	Mean	Standard Deviation
Trainer 1	110	8.6818	4.6580
Trainer 2	110	5.1091	2.4244
Recorded Sessions by Trainer 1	110	5.6364	3.3973
Overall	**330**	**6.4758**	**3.9303**

Trainer 1 recorded a substantially higher mean Technical Expertise score of **8.6818**. Trainer 2 recorded the lowest mean at **5.1091**, while the Recorded Sessions group recorded a mean of **5.6364**.

The descriptive results suggest that the difference between training conditions is considerably more pronounced for Technical Expertise than for Design Modelling.

7. Multivariate Analysis of Variance (MANOVA)

A one-way MANOVA was conducted to determine whether the three training conditions differed significantly when the two dependent variables were considered together.

The dependent variable vector consisted of Design Modelling and Technical Expertise. The multivariate tests produced statistically significant results across all four commonly reported MANOVA statistics:

Multivariate Test	Value	F	df	Significance
Pillai's Trace	0.173	15.442	4, 654	p < .001
Wilks' Lambda	0.828	16.117	4, 652	p < .001
Hotelling's Trace	0.207	16.791	4, 650	p < .001
Roy's Largest Root	0.202	33.084	2, 327	p < .001

The results consistently indicate a statistically significant multivariate effect of **Training_Group** on the combined dependent variables.

> **Primary Statistic (Wilks' Lambda):**
> $\Lambda = 0.828, F(4, 652) = 16.117, p < .001$

Therefore, the null hypothesis that the three training groups have equal mean vectors across Design Modelling and Technical Expertise is rejected. The MANOVA results provide evidence that the overall pattern of learning outcomes differs significantly across the three training conditions.

8. Follow-Up Univariate ANOVA

Because the overall MANOVA was statistically significant, follow-up univariate ANOVAs were examined to determine which individual dependent variables contributed to the overall multivariate effect.

8.1 Design Modelling

The univariate ANOVA for Design Modelling produced $F(2, 327) = 5.146, p = .006$. The result is statistically significant at the .05 significance level.

The partial eta squared effect size was $\eta_p^2 = .031$, indicating that approximately **3.1% of the variance** in Design Modelling scores is associated with the Training_Group factor within this model. The effect is relatively small compared with Technical Expertise.

8.2 Technical Expertise

The univariate ANOVA for Technical Expertise produced $F(2, 327) = 31.379, p < .001$. This result is highly statistically significant.

The partial eta squared effect size was $\eta_p^2 = .161$, indicating that approximately **16.1% of the variance** in Technical Expertise scores is associated with the Training_Group factor within this model.

The effect size is substantially larger than that observed for Design Modelling, indicating a considerably stronger association with Technical Expertise performance.

Outcome Measure	df	F Value	p-value	η_p^2	Effect Size
Design Modelling	2, 327	5.146	.006	.031	Small (3.1%)
Technical Expertise	2, 327	31.379	< .001	.161	Large (16.1%)

9. Estimated Marginal Means

The estimated marginal means provide adjusted group-level estimates for each dependent variable.

9.1 Design Modelling

Training Group	Mean	Standard Error	95% Confidence Interval
Trainer 1	6.191	0.187	5.822 – 6.560
Trainer 2	5.582	0.187	5.213 – 5.950
Recorded Sessions by Trainer 1	5.373	0.187	5.004 – 5.741

Trainer 1 recorded the highest estimated marginal mean. The difference between Trainer 1 and Recorded Sessions was larger than the difference between Trainer 1 and Trainer 2.

9.2 Technical Expertise

Training Group	Mean	Standard Error	95% Confidence Interval
Trainer 1	8.682	0.344	8.005 – 9.359
Trainer 2	5.109	0.344	4.432 – 5.786
Recorded Sessions by Trainer 1	5.636	0.344	4.959 – 6.314

Trainer 1 recorded a substantially higher estimated marginal mean than both Trainer 2 and Recorded Sessions.

10. Tukey HSD Post-Hoc Analysis

Tukey's Honestly Significant Difference (HSD) procedure was used to identify which specific pairs of training conditions differed significantly ($\alpha = .05$).

10.1 Design Modelling Pairwise Comparisons

- **Trainer 1 vs. Trainer 2:** Mean Difference = 0.6091, p = .057 (Not Statistically Significant). Although Trainer 1 had a higher observed mean, the evidence was insufficient to conclude that the two groups differed significantly.
- **Trainer 1 vs. Recorded Sessions:** Mean Difference = 0.8182, p = .006 (Statistically Significant). Participants in the Trainer 1 live training condition achieved significantly higher scores than those receiving Recorded Sessions.
- **Trainer 2 vs. Recorded Sessions:** Mean Difference = 0.2091, p = .710 (Not Statistically Significant). No statistically significant difference was observed.

10.2 Technical Expertise Pairwise Comparisons

- **Trainer 1 vs. Trainer 2:** Mean Difference = 3.5727, p < .001 (Highly Statistically Significant). Trainer 1 achieved substantially higher scores.
- **Trainer 1 vs. Recorded Sessions:** Mean Difference = 3.0455, p < .001 (Highly Statistically Significant). Live Trainer 1 training significantly outperformed Recorded Sessions.
- **Trainer 2 vs. Recorded Sessions:** Mean Difference = -0.5273, p = .525 (Not Statistically Significant). No significant difference found.

11. Homogeneous Subsets

11.1 Design Modelling

The Recorded Sessions group and Trainer 2 appeared together in Subset 1 (no significant difference). Trainer 2 also appeared in Subset 2 with Trainer 1, acting as a statistical bridge. Trainer 1 and Recorded Sessions were placed in separate subsets:

> **Design Modelling Pattern:** Trainer 1 > Trainer 2 > Recorded Sessions *(Only Trainer 1 vs. Recorded Sessions is statistically significant)*

11.2 Technical Expertise

Trainer 2 and Recorded Sessions appeared together in Subset 1. Trainer 1 appeared separately in Subset 2 as a significantly higher-performing group:

Technical Expertise Pattern: Trainer 1 > Recorded Sessions ≈ Trainer 2 *(Trainer 1 significantly outperforms both)*

12. Profile Plot Interpretation

Design Modelling Profile: Shows a gradual decline in observed performance from Trainer 1 (6.19) to Trainer 2 (5.58) and then to Recorded Sessions (5.37). The relatively small effect size indicates limited practical difference.

Technical Expertise Profile: Demonstrates a substantial visual separation. Trainer 1 (8.68) sits far above Trainer 2 (5.11) and Recorded Sessions (5.64), visually reinforcing the large effect size.

13. Key Findings

- **Finding 1 (Significant Multivariate Effect):** MANOVA demonstrated a statistically significant difference across conditions for combined learning outcomes.
- **Finding 2 (Stronger Effect for Technical Expertise):** Training condition explained 16.1% of variance in Technical Expertise ($\eta_p^2 = .161$) compared to 3.1% in Design Modelling ($\eta_p^2 = .031$).
- **Finding 3 (Peak Performance for Trainer 1):** Trainer 1 achieved the highest mean score in both Design Modelling (6.1909) and Technical Expertise (8.6818).
- **Finding 4 (Strong Technical Advantage):** Live Trainer 1 significantly outperformed Trainer 2 (+3.57 pts) and Recorded Sessions (+3.05 pts) in Technical Expertise (both $p < .001$).
- **Finding 5 (Live vs. Recorded Format):** Live instruction by Trainer 1 produced significantly higher results than Recorded Sessions by the same trainer across both outcomes.
- **Finding 6 (Trainer 2 Equals Recorded Sessions):** Trainer 2 and Recorded Sessions did not differ significantly in either outcome measure.

14. Practical Interpretation

From a training and development perspective, delivery method and instructor selection are important factors. The findings suggest live instruction may provide opportunities for interaction, feedback, and immediate clarification that are particularly valuable when building technical competencies.

Recorded training may not yield identical outcomes to live instruction, even when delivered by the same trainer. However, these results reflect observed associations rather than definitive evidence of direct causality.

15. Limitations

- **Observational Interpretation:** Results show associations rather than establishing strict causality.
- **Generalizability:** Findings based on N = 330 observations may not generalize to all corporate or academic environments.
- **Limited Outcome Measures:** Focused on two domain metrics; satisfaction, engagement, and long-term retention were unmeasured.
- **Confounding Variables:** Prior participant experience, motivation, and learning preferences were not independently controlled.
- **Statistical Assumptions:** Interpretations depend on standard MANOVA assumptions evaluated during SPSS workflow execution.

16. Overall Conclusion

The MANOVA analysis confirms that training delivery methods significantly affect combined learning outcomes, with the multivariate effect driven primarily by Technical Expertise.

Live training delivered by Trainer 1 produced the strongest learning outcomes overall, demonstrating statistically significant advantages in Technical Expertise over both Trainer 2 and Recorded Sessions. For technical competency development, interactive instructor-led training appears highly advantageous.

17. Evidence and Project Files

- `Data/manova (1).sav` — Original SPSS dataset.
- `Output/Output1manova.spv` — Original SPSS Viewer statistical output file.
- `Documentation/Screenshots/` — Output verification screenshots for factor tests, ANOVA tables, pairwise comparisons, and profile plots.

18. Learning and Attribution Note

This report was completed as part of an applied research learning project in statistical analysis using IBM SPSS Statistics. It demonstrates hands-on experience in MANOVA, univariate follow-up tests, post-hoc analysis, effect size interpretation, and data-driven technical reporting.

19. Project Status

> **Project Status**
>
> **Status: Completed – Portfolio Documentation**
> The MANOVA research project is fully documented and structured within a GitHub-ready project repository.